

Erika Sy

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EDUCATION

University of Wisconsin - Madison | **B.S. in Computer Science & Data Science (Honors)** **Minor: Mathematics & Business** Dec. 2026
Coursework - Honors Algorithms, Database Management, Computer Networks, Artificial Intelligence, Honors Computer Graphics

SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, SQL, C, R
Frameworks & Libraries: FastAPI, React, Next.js, TailwindCSS, TensorFlow, PyTorch
Tools & Platforms: Git, Docker, PostgreSQL, Docker, Redis, Neo4j, CHTC, Unreal Engine

EXPERIENCE

University of Wisconsin - Madison | **Undergraduate Research Assistant (Dr. Gleicher)** Sep. 2025 - Present

- Built an interactive **NLP** document exploration tool (Streamlit, Plotly) for **1,100+** research documents, enabling linked multi-view analysis across **t-SNE**, **UMAP**, and **PCA** projections with lasso/box selection.
- Designed a modular end-to-end text analytics pipeline in **Python** spanning preprocessing, embeddings, clustering, and dimensionality reduction across **7** configurable components.
- Optimized research iterations by implementing real-time K-Means re-clustering and a reproducible experimentation framework with **9+** integration tests, reducing pipeline latency from minutes to seconds.

University of Wisconsin - Madison | **Undergraduate Research Assistant (Dr. Hanna)** Sep. 2024 - Present

- Accelerated **RL** agent training by automating reward function optimization with **evolutionary algorithms** and parallel computing on **CHTC**.
- Discovered novel **robotic gait patterns** in simulation that demonstrated improvements in stability and traversal speed over baseline models.
- Built a **synthetic data pipeline** in Unreal Engine 4, generating **10,000+** labeled images to train and benchmark real-time computer vision models.

AbbVie | **Business Technology Solutions Intern** May 2025 - Aug. 2025

- Automated mapping of **200+** document workflows with a **Neo4j** graph database, establishing a single source of truth to eliminate manual tracking.
- Developed a **full-stack AI-powered** onboarding platform using **RAG**, **React**, and **Python** to centralize "tribal knowledge".
- Built a high-performance ETL parser in **Python** to extract and structure metadata from **100+** decentralized, non-standardized spreadsheets, replacing manual lookup with a searchable index and reducing data retrieval latency by **75%**.

NASA NPWEE | **Lead Software Engineer** Jan. 2025 - May 2025

- Directed a **4-person** team to architect a **full-stack** mission scheduling tool; owned the system design and stack selection (React, FastAPI, PostgreSQL).
- Managed the end-to-end Project Lifecycle via Agile sprints, translating mission requirements into a **15-page** technical **proposal**.
- Orchestrated containerized deployment using **Docker**, streamlining the integration of heuristic-based scheduling algorithms.

Snap Inc. | **Augmented Reality Development Extern** Aug. 2023 - May 2024

- Shipped **4** augmented reality lenses using Lens Studio and **JavaScript**.
- Optimized 3D asset overhead and scripted logic to maintain **60 FPS** across low-end devices while synchronizing multi-user state for thousands of active users.

PERSONAL PROJECTS

[Stride](#) | *FastAPI, SQLAlchemy, Next.js, React, Tailwind CSS* Oct. 2025 - Present

- Developed a **full-stack** social platform enabling users to share daily accomplishments and maintain engagement streaks with friends.
- Implemented real-time features using **WebSocket/Redis** for live feeds and **JWT** authentication for secure user sessions.
- Built a responsive frontend with **Next.js** and **TailwindCSS**; currently developing a **React** Native mobile app with push notifications.

[Path-Mind](#) | *Arduino, C++, Python, Q-Learning, PyTorch, Matplotlib* Mar. 2025 - Present

- Built an **autonomous maze-solving robot** from scratch, implementing sensor fusion and motor control in **C++** for real-time navigation.
- Trained a **Q-Learning** agent that discovered optimal paths through self-exploration, achieving a **23%** reduction in solve time vs. heuristic baselines.

[PlantDocNet](#) | *TensorFlow, Keras, MobileNetV2, scikit-learn, Python* Apr. 2025

- Trained a deep learning classifier using **MobileNetV2** on **20 K+** leaf images from a dataset, achieving **89.1%** validation accuracy across **16** disease classes. Assessed performance using **scikit-learn** metrics to analyze model robustness and class balance.
- Designed a full **ML pipeline** with image augmentation, data splitting, transfer learning, and early stopping to ensure stable convergence.

AWARDS

Code Your Dreams (Ideathon) | **Best Community Outreach Award** Apr. 2024

- Awarded** for designing an accessible STEM event calendar prototype to increase engagement for underrepresented groups.